

Introduction

- Converts long URLs into short, unique links.
- Redirects users from short URLs to original URLs.
- Designed for fast, scalable, and reliable URL redirection.

Objectives

- Minimize redirect latency.
- Support billions of URLs.
- Ensure high availability.
- Provide secure and reliable service.

Key Features

- URL shortening
- Custom aliases
- Link expiration
- Click analytics
- User authentication
- QR code generation (optional)

Design Goals

- High scalability
- Fault tolerance
- Fast cache-based lookups
- Reliable storage with backups
- Easy maintenance and monitoring

URL SHORTENER — SYSTEM DESIGN

